

GOKULA UDHAYAN V

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PROFESSIONAL SUMMARY

Highly motivated B.Tech Information Technology undergraduate (8.33 CGPA) with a strong technical foundation in full-stack SDE and Agentic AI applications. Proven experience engineering scalable web systems using React, Django, and MongoDB, alongside practical implementation of advanced RAG pipelines, dynamic Knowledge Tracing frameworks, and containerized deployments. Eager to contribute as an SDE intern or full-time engineer in innovation-driven product teams.

TECHNICAL SKILLS

Programming Languages	Python, C++, JavaScript (ES6+), SQL, PHP, HTML5, CSS3
Frontend & Backend	React, Django, REST APIs, JSON, Bootstrap
AI, ML & GenAI Frameworks	PyTorch, TensorFlow, LangChain, LangGraph, RAG, Agentic AI, BKT
Databases & Clouds	MongoDB, MySQL, Neo4j, Vercel, AWS (EC2, ECS, S3), Docker
Developer Tools & DevOps	Git, GitHub Actions (CI/CD), Docker, Postman, Linux, Streamlit

PROJECTS

AI-Powered Agentic Tutoring System | *React, Django, LangGraph, Neo4j, AWS ECS*

- Built a full-stack tutoring pipeline leveraging Graph Neural Networks (GNNs) for concept mapping and Bayesian Knowledge Tracing (BKT) engine in Python to track student conceptual mastery with **94.2% accuracy**.
- Optimized prerequisite graph traversals in Neo4j, reducing processing delay by **18%**; engineered **docker-based** multi-tier containerized deployments with a **CI/CD pipeline** targeting **AWS ECS** to maintain **90%+ test coverage**.

ONDC based Ecommerce platform| *Next.js, MySQL, Vercel, REST APIs* ondc-based-ecommerce-platform.vercel.app/

- Engineered multi-tenant, location-aware ONDC architecture** using Next.js App Router and Vercel Postgres; cut geolocal sorting to $O(1)$ complexity via spatial bounding-box indexing and cursor streaming.
- Designed high-concurrency product inventory engine** using optimistic UI and atomic SQL transactions; eliminated race conditions under heavy load and enforced single-store cart isolation for network compliance.

AI Smart Travel Planner | *Python, Groq API, Streamlit, DuckDuckGo API, Asyncio* tripplannercsv.streamlit.app

- Engineered a session-persistent route planning application on Streamlit combining LLM generation (Groq API) with real-time web crawlers (DuckDuckGo API) to output dynamically refined, structural holiday itineraries.
- Implemented asynchronous fetching routines utilizing Python's *asyncio* library, lowering concurrent network API search response times by **35%** to complete plan generation tasks under **2.5 seconds**.

LoRA-Tuned LLaMA Chatbot | *PyTorch, LoRA, LLaMA, Hugging Face* hf.space/device-expert

- Fine-tuned a LLaMA-7B parameter model using Parameter-Efficient Fine-Tuning (PEFT) and Low-Rank Adaptation (LoRA) constraints in PyTorch to automate conversational device diagnostics and interactive troubleshooting flows.
- Trained on a curated corpus of **1,000+ custom device-reasoning pairs** to hit an evaluation perplexity of **2.89**, compressing the active inference memory footprint by **65%** via 4-bit QLoRA to facilitate responsive, low-cost execution on constrained server runtimes.

PUBLICATIONS

Book Chapter Citation

Gokula Udhayan, V., Mahaeshwari, K., & Vinoth Kumar, N. (2024). Advancement of ML in smart house. In S. Mahajan, K. Joshi, A. K. Pandit, & N. Pathak (Eds.), *Integrating Metaheuristics in Computer Vision for Real-World Optimization Problems* (pp. 207–224). Wiley-Scrivener.

CERTIFICATIONS

Introduction to LangGraph (LangChain Academy, 2026); Generative AI: Foundation Models and Platforms (IBM, 2025); Supervised Machine Learning: Regression (IBM, 2025); Automating Real-World Tasks with Python (Google, 2025); OpenAI GPT-3 for Developers (Infosys, 2024); UiPath Orchestrator and Capstone Automation (UiPath Academy, 2025).

EDUCATION

Bachelor of Technology in Information Technology	(Aug 2022 – May 2026)
Kumaraguru College of Technology, Coimbatore, Tamil Nadu, India	(CGPA: 8.33/10.0)
Higher Secondary Education (HSE)	(June 2020 – May 2022)
K.V.S Matriculation Higher Secondary School, Virudhunagar, India	(Marks: 95%)